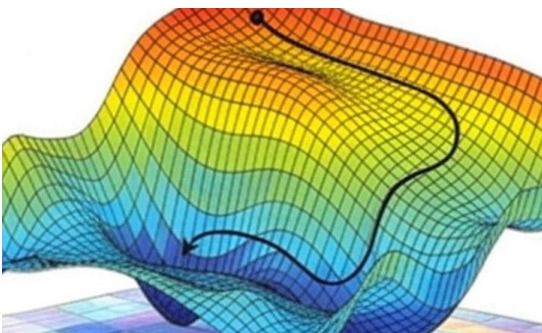


1 Introduction

Large Scale Problem of dimension n but iterations $\ll n$ desired

Convex One of the only problem classes that are “solvable”

Optimization of objective function f with decision variable x considering feasible set $\mathcal{C} = \{\xi \in \mathbb{R}^n : g(\xi) \leq 0, h(\xi) = 0\}$



Local minimum x^* if $\exists \epsilon > 0$ s.t. $f(x^*) \leq f(x), \forall x \in \mathcal{C} \cap B_\epsilon(x^*), B_\epsilon(x^*) := \{\xi \in \mathbb{R}^n : |\xi - x^*| < \epsilon\}$

Proposition 1. f (lower-semi-)continuous, radially unbounded, $\mathcal{C}$ closed $\Rightarrow \exists \min_{x \in \mathcal{C}} f(x)$ and $x^* \in \operatorname{argmin}_{x \in \mathcal{C}} f(x)$

Definition 1 (Lipschitz continuity). $q : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is Lipschitz with constant L if: $|q(x) - q(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$

f is **Lipschitz** (Lip) with constant $L \Leftrightarrow |\nabla f(x)|_2 \leq L$

OP class $\mathcal{P}$ with $\mathcal{C} = [0, 1]^n$, f is l^∞ -Lipschitz with constant L

Proposition 2. For any algorithm $\exists$ problem in $\mathcal{P}$, s.t. achieving $|f(x_N) - f(x)| < \epsilon$ requires $N \geq (\lfloor \frac{L}{2\epsilon} \rfloor)^n - 1$

Definition 2. OP convex if, f and g_i convex functions, h affine.

Definition 3. $q : \mathbb{R}^n \rightarrow \mathbb{R}$ convex (affine) if $\forall x, y \in \mathbb{R}^n$

$$q(\theta x + (1-\theta)y) \leq \theta q(x) + (1-\theta)q(y) \quad \forall \theta \in [0, 1]$$

Proposition 3. OP convex $\Rightarrow$ local minimum = global minimum

2 Convex Optimization Problem

Definition 4 (Convex Set). A set $\mathcal{C}$ is convex if and only if

$$\theta x + (1 - \theta)y \in \mathcal{C} \forall x, y \in \mathcal{C}, \quad \forall \theta \in [0, 1]$$

(hyperplane || half-space) $\{x \in \mathbb{R}^n \mid a^\top x (=||\leq) b\}$

polyhedra $\{x \in \mathbb{R}^n \mid A^{q \times n} x \preceq b^{q \times 1}, C^{r \times n} x = d^{r \times 1}\}$

Operations that preserve convexity (sets)

Intersection $\mathcal{C}_1, \mathcal{C}_2$ cv $\Rightarrow \mathcal{C}_1 \cap \mathcal{C}_2$ convex (**cv**)

Image under affine map $\mathcal{C} \subseteq \mathbb{R}^n$ cv $\Rightarrow \{Ax + b \mid x \in \mathcal{C}\}$ cv

Inverse loaM $\mathcal{C} \subseteq \mathbb{R}^m$ cv $\Rightarrow \{x \in \mathbb{R}^n \mid Ax + b \in \mathcal{C}\}$ cv

Separating Hyperplane Theorem

Theorem 1. $\mathcal{C} \subseteq \mathbb{R}^n$ non-empty closed (**cl**) convex set, $y \notin \mathcal{C} \rightarrow \exists a \neq 0, b \in \mathbb{R}$ s.t. $a^\top x + b < a^\top y + b, \forall x \in \mathcal{C}$

Corollary 1. $\mathcal{C}_{\text{cl,cv}}$: intersection of cl half-spaces that contain $\mathcal{C}$

Support function

Idea represent any cl,cv set by its supporting hyperplanes

$$\sigma_{\mathcal{C}}(a) = \sup_{x \in \mathcal{C}} a^\top x \quad \text{if known, one can construct}$$

$$\mathcal{C} = \bigcap_{a \in \mathbb{R}^n} \{x \in \mathbb{R}^n \mid a^\top x - \sigma_{\mathcal{C}}(a) \leq 0\}$$

$$= \{x \in \mathbb{R}^n \mid \sup_{a \in \mathbb{R}^n} a^\top x - \sigma_{\mathcal{C}}(a) \leq 0\}$$

Definition 5. $f : \mathbb{R}^n \rightarrow \mathbb{R}$ cv $\Leftrightarrow$ epigraph of f is cv set

$$\operatorname{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}$$

$\rightarrow$ this provides a link between convex sets and functions

Operations that preserve convexity (functions)

- the point wise maximum of convex functions is convex
- the sum of convex functions is convex
- $f(Ax + b)$ is convex if f is convex

Check Convexity f is convex if it is composition of simple convex function with convexity preserving operations or if

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, $\partial^2 f / \partial x^2 \succeq 0 \forall x \in \mathbb{R}^n$

$g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(t) = f(x + tv)$ convex in $t \forall x, v \in \mathbb{R}^n \rightarrow f$ convex (restriction to a line)

Extended real numbers $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$

Indicator function $\psi_C(x) := \begin{cases} +\infty & \text{if } x \notin C \\ 0 & \text{if } x \in C \end{cases}$

$\rightarrow$ this provides another link between convex sets and functions

We can write $\min_{x \in C} f(x)$ as $\min_{x \in \mathbb{R}^n} f(x) + \psi_C(x)$

Definition 6 (3). $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is called proper if f is bounded below and if $\exists x \in \mathbb{R}^n$ s.t. $f(x) < \infty$

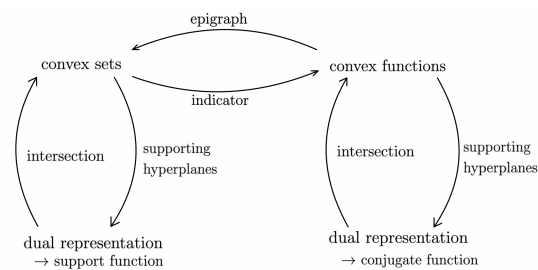
Definition 7 (Legendre Transformation). The **conjugate function** of $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is defined as $f^*(y) = \sup_{x \in \mathbb{R}^n} y^T x - f(x)$

Concave $\nabla_x^2 f^* \preceq 0 \Rightarrow$ maximizer of sup satisfies $\nabla_x f^* = 0$

Theorem 2 (Conjugate of Conjugate). $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$

(i) f proper, cv, epi(f) closed $\Rightarrow f^{**} = f$

(ii) $f(x) \geq f^{**}(x), \forall x \in \mathbb{R}^n$



3 KKT and Lagrange Duality

We consider $f^* = \inf_{x \in \mathbb{R}^n} f(x)$ s.t. $g(x) \leq 0, h(x) = 0$ (1)

Lagrange $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T h(x)$

Dual Function $d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu)$

Proposition 4 (Weak Duality). $d(\lambda, \nu) \leq f^*, \forall \lambda \geq 0, \nu \in \mathbb{R}^h$

Definition 8 (Constraint qualification). C convex, **Slater's Condition** holds if $\exists \hat{x} \in \mathbb{R}^n$ s.t. $h(\hat{x}) = 0$ and $g(\hat{x}) < 0$

Proposition 5 (Strong Duality). If Slater's condition holds and (1) is convex $\Rightarrow \exists \lambda \geq 0, \nu \in \mathbb{R}^{n_h}$ s.t. $d(\lambda, \nu) = f^*$

KKT (Karush-Kuhn-Tucker) Conditions

Theorem 3 (KKT Conditions). Slater's condition holds and (1) is convex $\rightarrow x^* \in \mathbb{R}^n$ is a minimizer of the primal (1) and $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^{n_g} \times \mathbb{R}^{n_h}$ is a maximizer of the dual $\Leftrightarrow$

- | | |
|---|---------------------------------|
| $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$ | KKT-1 (Stationary Lagrangian) |
| $g(x^*) \leq 0, h(x^*) = 0$ | KKT-2 (primal feasibility) |
| $\lambda^* \geq 0, \nu^* \in \mathbb{R}^{n_h}$ | KKT-3 (dual feasibility) |
| $\lambda^{*T} g(x^*) = 0 = \nu^{*T} h(x^*)$ | KKT-4 (complementary slackness) |

In addition we have: $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^{n_h}} q(\lambda, \nu) = \inf_{x \in C} f(x)$

Remark Without Slater, KKT1-4 still implies x^* minimizes (1) and λ, ν maximizes dual, but the converse is no longer true. There can be primal-minimizer/dual-maximizer not satisfy KKT.

Subdifferential

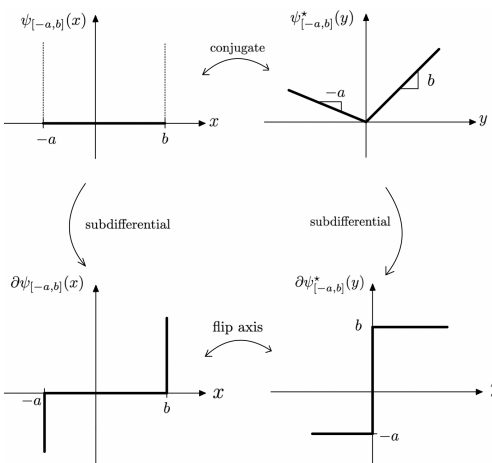
For cv f we have $f(x) \geq f(\bar{x}) + \nabla f(\bar{x})^T (x - \bar{x}), \forall x, \bar{x} \in \mathbb{R}^n$

Definition 9. $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ cv, the **subdifferential of f at $\bar{x}$** is:

$\partial f(\bar{x}) := \{\lambda \in \mathbb{R}^n \mid f(x) \geq f(\bar{x}) + \lambda^T (x - \bar{x}), \forall x \in \mathbb{R}^n\}$

Proposition 6. f (like D9), $x^* \in \operatorname{argmin}_x f(x) \Leftrightarrow 0 \in \partial f(x^*)$

Proposition 7 (Relation to conjugate functions). For convex f with epi(f) closed: $y \in \partial f(x) \Leftrightarrow x \in \partial f^*(y)$



4 Convex Optimization Problems

Optimal value $f^* = \inf\{f(x) \mid g_i(x) \leq 0, h_j = 0\}$
 $f^* = +\infty$ OP is infeasible, $f^* = -\infty$ OP is unbound below

Feasibility Problem

Special case $f(x) = 0, \forall x \Leftrightarrow \min_s \text{ s.t. } g_i(x) \leq s, h_j(x) = 0$

Linear Programming minimize $c^T x$ s.t. $Ax - b \geq 0, x \geq 0$

Step 1: $\mathcal{L}(x, \lambda_1, \lambda_2) = c^T x - \lambda_1^T (Ax - b) - \lambda_2^T x, \lambda_i \geq 0$

Step 2: $\inf_{x \in \mathbb{R}^n} \mathcal{L} = \lambda_1^T b$, if $c - A^T \lambda_1 - \lambda_2 = 0$, else $-\infty$

Step 3: Dual, maximize $b^T \lambda$ s.t. $c - A^T \lambda \geq 0, \lambda \geq 0$ (again LP)

Proposition 8. The optimal solution of a linear program (if it exists) lies always on the boundary of the feasible set and there exists an optimal solution that is a vertex of the feasible set.

Quadratic Programming convex if $P = P^T$ positiv semi-definite
 minimize $\frac{1}{2} x^T P x + q^T x$ s.t. $Gx \leq h, Ax = b$

Second-Order Cone Program

minimize $f^T x$ s.t. $|A_i x + b| \leq c_i^T x + d_i, Fx = g$

Second-order cone $C_{n+1} = \{(x, t) \mid x \in \mathbb{R}^n, t \in \mathbb{R}, |x| \leq t\}$

$|A_i x + b| \leq c_i^T x + d_i \Leftrightarrow (A_i x + b, c_i^T x + d_i) \in C_{n+1}$

Semi-Definite Programming with symmetric F_i, X, A_i

minimize $c^T x$ s.t. $\sum_{i=1}^n x_i F_i + G \preceq 0, Ax = b$

Standard form minimize $\text{tr}(CX)$ s.t. $X \succeq 0, \text{tr}(A_i X) = b_i$
 $\text{tr}(CX) = \sum_{i=1}^n \sum_{j=1}^n C_{ij} X_{ij}, C \in \mathbb{R}^{n \times n}, i = 1, \dots, m$

$\text{LP} \subset \text{QP} \subset \text{QCQP (Quadratically Constrained QP)} \subset \text{SOCP} \subset \text{SDP}$

5 Gradient methods - Part I

Definition 10 (smoothness). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth (**L -sm**) if $\nabla f(x)$ satisfies $|\nabla f(x) - \nabla f(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$

Taylor $\rightarrow f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} |x - y|^2$

Definition 11 (strong convexity). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex (**μ -scv**) if $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} |x - y|^2$

How to find μ/L , Spectra of Hessian $\nabla^2 f$, min/max eigenvalue

Gradient Descent

$x_{k+1} = x_k - T \nabla f(x_k)$ for $k = (k_0, \dots, k_N)$ given x_0, T

Assume $f(x) = c_0 + b^T x + \frac{1}{2} x^T H x, H \succ 0 \Rightarrow H x^* = -b$

$x_{k+1} - x^* = x_k - x^* - T(b + H x_k) = (I - TH)(x_k - x^*)$

Convergence given by eigenvalues of $I - TH$, use $H = U \Lambda U^T$

$x_N - x^* = U(I - T \Lambda)^N U^T (x_0 - x^*) \rightarrow \text{conv-rate } 1 - T \lambda_i$

$f: L\text{-sm}, \mu\text{-scv} \rightarrow \mu \leq \min \lambda_i, \max \lambda_i \leq L, \rightarrow \text{conv-rate } \rho(T)$
 $=: \max_{\mu \leq h \leq L} |1 - Th| \rightarrow |x_N - x^*| \leq \rho(T)^N |x_0 - x^*|$

$T^* = \frac{2}{L+\mu}$, with **condition number** $\kappa := \frac{L}{\mu}$ and $1 - \xi \leq e^{-\xi}$

$\rho(T^*) = \frac{L-\mu}{L+\mu} = \frac{\kappa-1}{\kappa+1} = (1 - \frac{2}{\kappa+1}) \leq e^{-\frac{2}{\kappa+1}} \rightarrow \text{algebraic}$

complexity $N \geq \frac{\kappa+1}{2} \ln(\frac{|x_0 - x^*|}{\epsilon})$ to achieve $|x_N - x^*| \leq \epsilon$

Momentum-based methods

$$q_{k+1} = q_k + \frac{1}{L} p_{k+1}$$

$$p_{k+1} = (1 - 2dT)p_k - T \nabla f(q_k + \beta p_k)/L$$

Nesterov accelerated gradient $T = 1, d = \frac{1}{\sqrt{k+1}}, \beta = \frac{\sqrt{k}-1}{\sqrt{k+1}}$

Heavy Ball tuned on quadratic $T = \frac{2\sqrt{\kappa}}{\sqrt{k+1}}, d = \frac{1}{\sqrt{k+1}}, \beta = 0$

$$C_{\text{Nesterov}}(1 - \frac{1}{\sqrt{\kappa}})^N \approx \frac{|q_N - q^*|}{|q_0 - q^*|} \approx C_{\text{HeavyBall}}(1 - \frac{2}{\sqrt{\kappa+1}})^N$$

Theorem 4. f : L -sm, μ -scv $\rightarrow$ Nesterov's method satisfies:

$$|q_N - q^*| \leq \sqrt{\kappa + 1}(1 - 1/\sqrt{\kappa})^{N/2}|q_0 - q^*|$$

$$f(q_N) - f^* \leq \frac{L + \mu}{2}(1 - 1/\sqrt{\kappa})^N |q_0 - q^*|^2$$

Requires $N \geq 2\sqrt{\kappa} \ln(\frac{|q_0 - q^*|}{\epsilon})$ to achieve $|x_N - x^*| \leq \epsilon$

Theorem 5. For any first-order method $\exists f : \mathbb{R}^\infty \rightarrow \mathbb{R}, \mu$ -scv, L -sm, s.t. $|x_k - x^*| \geq (1 - \frac{2}{\sqrt{\kappa+1}})^k |x_0 - x^*| \forall k \geq 0$

Line search optimal step $\nu_t^* = \operatorname{argmin}_{\nu \in \mathbb{R}} f(x_t - \nu \nabla f(x_t))$

6 Gradient Methods - Part II

Definition 12. $\operatorname{prox}_C(x) = \operatorname{argmin}_{y \in C} \frac{1}{2}|x - y|^2$ $C \subset \mathbb{R}^n$

Lemma 1. cl, cv $C \subset \mathbb{R}^n \rightarrow |\operatorname{prox}_C(x) - \operatorname{prox}_C(y)| \leq |x - y|$
 $\leftarrow |\operatorname{prox}_C(x) - \operatorname{prox}_C(y)|^2 \leq (\operatorname{prox}_C(x) - \operatorname{prox}_C(y))^T (x - y)$

Projected Gradient Descent

$x_{k+1} = \operatorname{prox}_C(x_k - T \nabla f(x_k))$, for $x_0, k_{0..N}, T \in (0, 2/L)$

Proposition 9. f : L -sm, μ -scv $\rightarrow$ projected GD with $T = \frac{2}{L+\mu}$ satisfies $|x_N - x^*| \leq |x_0 - x^*|(1 - \frac{2}{\kappa+1})^N$ (κ still $\frac{L}{\mu}$)

Lemma 2. $f : \mathbb{R}^n \rightarrow \mathbb{R}, L$ -sm, cv $\rightarrow \tilde{f}$ strongly-cv $\hat{f}(x) = f(x) + \frac{\mu}{2}|x - x_0|^2$ and $|\tilde{x}^* - x_0| \leq |x^* - x_0|$

and $f(x) - f(x^*) \leq \tilde{f}(x) - \tilde{f}(\tilde{x}^*) + \frac{\mu}{2}|x^* - x_0|^2, \mu > 0$

$\rightarrow$ from here one can apply GD or Nesterov, which results in:

$f(x_N) - f(x_0) \leq \epsilon$ after $N \sim L|x^* - x_0|^2/\epsilon$ iterations

Proposition 10 (Subgradient Method). $C_{cl, cv}$ contained in B_R

$x_{k+1} = \operatorname{prox}_C(x_k - T g_k), g_k \in \partial f(x_k), T = R/(L_f \sqrt{N})$

$\Rightarrow x_{0..N-1}$ satisfy $f(\frac{1}{N} \sum_{k=0}^{N-1}) - f(x^*) \leq \frac{RL_f}{\sqrt{N}}$

f	Method	$N \sim$	Optimal
μ -sc, L -sm	GD	$\kappa \ln(1/\epsilon)$	No
	NAG	$\sqrt{\kappa} \ln(1/\epsilon)$	Yes
L -smooth	GD	$1/\epsilon$	No
	NAG varying T	$1/\sqrt{\epsilon}$	Yes
L -Lip, cmp set	Subgradient	$1/\epsilon^2$	Yes

Example project a on l_1 -B: $\min_{x \in \mathbb{R}^n} \frac{1}{2}|x - a|_2^2$ s.t. $|x|_1 \leq 1$

$$\mathcal{L}(x, \lambda) = \frac{1}{2}|x - a|_2^2 + \lambda|x|_1 - \lambda$$

$$= \sum_{k=1}^n (\frac{1}{2}(x_k - a_k)^2 + \lambda|x|_1) - \lambda$$

$$d_k(\lambda) = \inf_{x_k \in \mathbb{R}} \phi(x_k) = \inf_{x_k \in \mathbb{R}} (\frac{1}{2}(x_k - a_k)^2 + \lambda|x|_1)$$

$$\partial \phi(x_k) = x_k - a_k + \lambda s, s(x_k = 0 : [1-1], else : sign(x_k))$$

$$d_k(\lambda) = \{\lambda \leq |a_k| : \frac{1}{2}\lambda^2 + \lambda|a_k|, else : \frac{1}{2}a_k^2\},$$

$$\nabla d_k(\lambda) = \max\{|a_k| - \lambda, 0\}$$

$$\text{Dual: } \max d(\lambda) \text{ s.t. } \lambda \geq 0 \text{ with } d(\lambda) = \sum_{n=1}^k d_k(\lambda) - \lambda$$

$$x_k = \{\lambda \geq |a_k| : 0; |a_k| > 0 : a_k - \lambda, |a_k| < 0 : a_k + \lambda\}$$

7 Stochastic Gradient Descent (SGD)

Stochastic Gradient $(\tilde{y}_i - \phi(\tilde{x}_i; \theta))^T \frac{\partial \phi}{\partial \theta} \big|_{\tilde{x}_i; \theta}, i \sim \{1, \dots, m\}$

Problem formulation: $\min_{x \in \mathbb{R}^n} F(x) = \min_{x \in \mathbb{R}^n} \mathbb{E}[f(x, \xi)]$

$$\mathbb{E}_\xi[f(x, \xi)] = \begin{cases} \int_{\mathbb{R}^q} f(x, \bar{\xi}) p_\xi(\bar{\xi}) d\bar{\xi} & \text{continuous Random V} \\ \sum_{\bar{\xi}} f(x, \bar{\xi}) p_\xi(\bar{\xi}) & \text{discrete R Variable} \end{cases}$$

$\text{Var}_\xi[g(x, \xi)] = \mathbb{E}_\xi[|g(x, \xi)|^2] - |\nabla F(x)|^2$ Step 1: $\xi_k \leftarrow$ generate realization of ξ

Step 2: $x_{k+1} = x_k - T_k g(x_k, \xi_k)$, step size T_k , SG $g(\cdot)$

$$\nabla_x f(x, \bar{\xi}), \bar{\xi} \sim p_\epsilon \text{ or } \frac{1}{n_{mb}} \sum_{i=1}^{n_{mb}} \nabla_x f(x, \bar{\xi}_i), \xi_i \sim p_\epsilon$$

$\Rightarrow$ The iterate x_k is now a random variable!

Assumptions on $F(x)$ and $g(x, \xi)$

A1 $F(x)$ is bounded below, ensures $\exists \min_x F(x)$ for $F : L$ -sm

A2 $\mathbb{E}_\xi[g(x, \xi)] = \nabla F(x), \forall x \in \mathbb{R}^n$, ensures SG unbiased.

A3 $\exists M, M_v \geq 0$ s.t. $\text{Var}_\xi[g(x, \xi)] \leq M + M_v|\nabla F(x)|^2$
 $\forall x \in \mathbb{R}^n$, ensures that variance is bounded.

Proposition 11. F μ -scv L -sm, SGD const. $T < \frac{1}{L(M_v+1)}$

$$\mathbb{E}[F(x_k)] - F(x^\star) \leq \frac{TLM}{2\mu} + (1 - T\mu)^k(F(x_0) - F(x^\star))$$

$$T = \frac{\ln(N)}{\mu N} \rightarrow N \sim \left(\frac{LM}{2\mu^2} + F(x_0) - F(x^\star)\right) / \epsilon$$

to ensure $\mathbb{E}[F(x_N)] - F(x^\star) \leq \epsilon$

$$(1 - T\mu)^N \leq e^{-T\mu N} \text{ this in EQ}$$

The role of mini batches $M \rightarrow M/n_{mb}, M_v \rightarrow M_v/n_{mb}$
 Same analysis holds, But run SGD with T/nmb to get same result... Advantage in computation if paralellization possible!

Non-(Strongly-)Convex Functions

Proposition 12. F, L -sm, SGD with $T \leq \frac{1}{L(1+M_v)}$ achieves

$$\mathbb{E}[\sum_{k=0}^{N-1} |\nabla F(x_k)|^2] \leq NTLM + \frac{2}{T}(F(x_0) - F_{\inf})$$

$$F_{\inf} = \inf_{x \in \mathbb{R}^n} F(x)$$

SGD Table (1 and 3 optimal, except for finite-sum minimization)

F	Criterion $\leq \epsilon$	$N \sim$	$T_k \sim$
μ -sc L -sm	$\mathbb{E}[F(\bar{x}_N)] - F(x_0)$	$1/\epsilon$	$1/k$
L -sm	$\mathbb{E}[\frac{1}{N} \sum_{k=0}^{N-1} \nabla F(x_k) ^2]$	$1/\epsilon^2$	$1/\sqrt{k}$
Lip, cv	$\mathbb{E}[F(\bar{x}_N)] - F(x_0)$	$1/\epsilon^2$	$1/\sqrt{k}$

8 ADMM

Parallelization $\min_{x \in \mathbb{R}^n} \sum_{i=1}^m f_i(x_i)$ s.t. $x_1 = \dots = x_m$

Dual ascent

Consider: $\min_{x \in \mathbb{R}^n} f(x)$ s.t. $Ax = b, A \in \mathbb{R}^{m \times n}$

Derive dual: $\mathcal{L}(x, \lambda) = f(x) + \lambda^\top Ax - \lambda^\top b$

$$\inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) = - \underbrace{\sup_{x \in \mathbb{R}^n} \{(-\lambda^\top A)x - f(x)\}}_{-f^\star(-A^\top \lambda)} - \lambda^\top b$$

Dual can be stated as: $\sup_{\lambda \in \mathbb{R}^m} \underbrace{-f^\star(-A^\top \lambda) - \lambda^\top b}_{:=d(\lambda)}$

Subgradient of d given by: $\partial d(\lambda) = A\partial f^\star(-A^\top \lambda) - b$

Recall $v \in \partial f^\star(u) \Leftrightarrow u \in \partial f(v)$ which means that the optimizer in $-\sup_{x \in \mathbb{R}^n} \{(-\lambda^\top A)x - f(x)\}$

satisfies: $-A^\top \lambda \in \partial f(x^\star) \Leftrightarrow x^\star \in \partial f^\star(-A^\top \lambda)$

As a Result, the subgradient $\partial d(\lambda)$ can be expressed via

$$\partial d(\lambda) = Ax - b, \text{ where } x \in \operatorname{argmin}_{\hat{x} \in \mathbb{R}^n} \{f(\hat{x}) + \hat{x}^\top A^\top \lambda\}$$

Dual Subgradient Method

$$x_k \in \operatorname{argmin}_{\hat{x} \in \mathbb{R}^n} \{f(\hat{x}) + \hat{x}^\top A^\top \lambda_k\}$$

$$\lambda_{k+1} = \lambda_k + T_k(Ax_k - b), \quad T_k > 0$$

Example $f(x) = \sum_{i=1}^m f_i(x_i)$ with $Ax = b$

$x = (x_1, \dots, x_n)$ and $A = [A_1, \dots, A_m]$

Dual subgradient becomes

$$x_{k_i} \in \operatorname{argmin}_{\hat{x}_i} \{f_i(\hat{x}_i) + \lambda_k^\top A_i \hat{x}_i\} \text{ (local minimization)}$$

$$\lambda_{k+1} = \lambda_k + T_k(\sum_{i=1}^m A_i x_{k_i} - b) \text{ (broadcasting)}$$

Proposition 13. f convex with closed epigraph, f is μ -strongly convex if and only if $f^\star$ is $1/\mu$ -smooth.

Derive ADMM

Idea: $\min_{x \in \mathbb{R}^n} f(x) + \frac{\rho}{2} |Ax - b|^2 \quad \text{s.t. } Ax = b \text{ with } \rho > 0$

Leads to this **Augmented Lagrangian**

$$x_k = \operatorname{argmin}_{x \in \mathbb{R}^n} f(x) + \lambda_k^\top Ax + \frac{\rho}{2} |Ax - b|^2$$

$$\lambda_{k+1} = \lambda_k + T_k(Ax_k - b) \quad (\text{typically } T_k = \rho)$$

Advantage Improved convergence properties even if f non-conv

Disadvantage Loss of decomposability/parallelization due to augmentation with quadratic term.

This motivates ADMM which tries to combine the best of both worlds. (Well conditioned minimization and parallelization)

Consider: $\min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} f(x) + g(z) \text{ s.t. } Ax + Bz = c$

Augmented Objective: $\min f(x) + g(z) + \frac{\rho}{2} |Ax + Bz - c|^2$

Augmented Lagrangian: objective + $\lambda^\top (Ax + Bz - c)$

Alternating Direction Method of Multipliers

$$x_k = \operatorname{argmin}_{x \in \mathbb{R}^n} \mathcal{L}_p(x, z_{k-1}, \lambda_k)$$

$$z_k = \operatorname{argmin}_{z \in \mathbb{R}^m} \mathcal{L}_p(x_k, z, \lambda_k)$$

$$\lambda_{k+1} = \lambda_k + \rho(Ax_k + Bz_k - c)$$

Examples

$\min_{x \in \mathbb{R}^n} |Ax - b|_1 \rightarrow \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} |z|_1 \text{ s.t. } Ax - z = b$

LP $\min_{x \in \mathbb{R}^n} c^\top x \text{ s.t. } Ax = b, x \geq 0$

$x_0 \in \mathbb{R}^n$ particular solution to $Ax_0 = b$

$Q \in \mathbb{R}^{n \times (n-m)}$ whose columns span null space of A .

Any solution x to $Ax = b$ can be represented by $x = x_0 + Qw$

$\rightarrow \min_{w \in \mathbb{R}^{n-m}} c^\top Qw + c^\top x_0 \text{ s.t. } x_0 + Qw \geq 0$

$\rightarrow \min_{w, z} c^\top Qw + c^\top x_0 \text{ s.t. } x_0 + Qw - z = 0, z \geq 0$

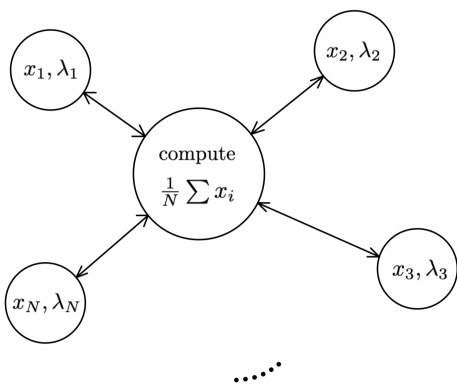
$\rightarrow \min_{w, z} c^\top Qw + c^\top x_0 + \psi_Z(z) \text{ s.t. } x_0 + Qw - z = 0$
($Z = \{z \in \mathbb{R}^n | z \geq 0\}$)

UPDATE RULES

9 Distributed optimization with ADMM

Goal Solve s.t. each term can be handled by its own processor.

$$\min_{x_1, \dots, x_N, z \in \mathbb{R}^n} \sum_{i=1}^N f_i(x_i) \quad \text{s.t. } x_i = z \quad (f_i \text{ convex}) \quad (2)$$



(Distributed ADMM)

Global Consensus Problem

Step 1: Augmented Lagrangian to solve (2) with ADMM.

$$\begin{aligned}\mathcal{L}_p(x_i, \dots, \lambda_i) &= \sum_{i=1}^N f_i(x_i) + \lambda_i^\top (x_i - z) + \frac{\rho}{2} |x_i - z|^2 \\ &= \sum_{i=1}^N f_i(x_i) + \frac{\rho}{2} |x_i - z + \frac{1}{\rho} \lambda_i|^2 - \frac{1}{2\rho} |\lambda_i|^2\end{aligned}$$

Step 2: Formulate ADMM

$$\begin{aligned}x_i^{k+1} &= \underset{x_i \in \mathbb{R}^n}{\operatorname{argmin}} f_i(x_i) + \frac{\rho}{2} |x_i - z^k + \frac{1}{\rho} \lambda_i^k|^2 \\ z^{k+1} &= \underset{z \in \mathbb{R}^n}{\operatorname{argmin}} \frac{\rho}{2} \sum_{i=1}^N |x_i^{k+1} - z + \frac{1}{\rho} \lambda_i^k|^2 \\ &= \frac{1}{N} \sum_{i=1}^N (x_i^{k+1} + \frac{1}{\rho} \lambda_i^k) \\ \lambda_i^{k+1} &= \lambda_i^k + \rho (x_i^{k+1} - z^{k+1})\end{aligned}$$

FURTHER REFORMULATIONS...

Sharing Problem

$$\min_{x_1, \dots, x_N \in \mathbb{R}^n} \sum_{i=1}^N f_i(x_i) + g\left(\sum_{i=1}^N x_i\right) \quad (3)$$

- copy all the variables $x_i = z_i$
- formulate augmented Lagrangian
- state ADMM dynamics

Optimization over Graphs

$g = (V, E)$ undirected graph with vertices V and edges E

$$\begin{aligned}\min_{x \in \mathbb{R}^n} \sum_{i \in V} f_i(x) &\Rightarrow \min_{x_i \in |V|, z_i \in |E|} \sum_{i \in V} f_i(x_i) \\ \text{s.t. } x_i &= z_{ij}, x_j = z_{ij} \quad \forall (i, j) \in E\end{aligned}$$

Step 1: Augmented Lagrangian

Step 2: Form the Algorithm

ALGORITHM

$$\begin{aligned}x_i^{k+1} &= \arg \min_{x_i \in \mathbb{R}^n} f_i(x_i) + \frac{d_i}{2} \left| x_i - \frac{1}{2} (x_i^k - \bar{x}_i^k) + \frac{1}{\rho} p_i^k \right|^2 \\ \bar{x}_i^{k+1} &= \frac{1}{d_i} \sum_{j \in \mathcal{N}_i} x_j^{k+1}, \\ p_i^{k+1} &= p_i^k + \frac{\rho}{2} (x_i^{k+1} - \bar{x}_i^{k+1}),\end{aligned}$$

10 Signal denoising and regression

Linear equation $y = Ax, y \in \mathbb{R}^n, x \in \mathbb{R}^m, A \in \mathbb{R}^{m \times n}$

classic setting $m \gg n$ | **modern** setting $m \ll n$ or $m \sim n$

Classic setting with outliers

l_2 -norm: $\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2$ results in heavy impact for outliers

l_1 -norm: $\min_{x \in \mathbb{R}^n} \|Ax - y\|_1$ can be reformulated and solved as

LP: $\min_{z \in \mathbb{R}^m} z^\top \mathbf{1}$ s.t. $-z \leq Ax - y \leq z, 0 \leq z$

$$\text{Combined: } \phi_{\text{Hub}}(u) = \begin{cases} u^2 & \text{if } |u| \leq M \\ 2Mu - M^2 & \text{if } |u| > M \end{cases}$$

Modern setting

$Ax = y$ infinite many solutions $\rightarrow$ add regularizer to find best

Tikhonov regression: $\min_{x \in \mathbb{R}^n} |Ax - y|_2^2 + \lambda |x|_2^2$

Least Absolute Shrinkage and Selection Operator

LASSO: $\min_{x \in \mathbb{R}^n} |Ax - y|_2^2 + \lambda |x|_1$

$\Leftrightarrow \min_{x \in \mathbb{R}^n} |Ax - y|_2^2$ s.t. $|x|_1 \leq c$ results in sparse solution

11 Classification

Setup Dataset with pairs of $(\tilde{x}_i, \tilde{y}_i)$, $i = 1, \dots, N$ with data $\tilde{x}_i \in \mathbb{R}^n$ and class $\tilde{y}_i \in \{1, 2, \dots, K\}$

Naive approach, linear regression

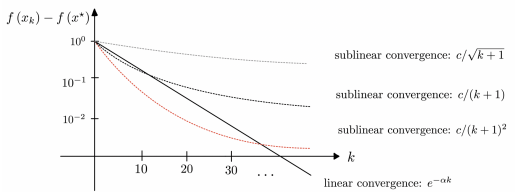
Slightly improved with probabilistic approach

Linear Discriminant Analysis

SVM aims to maximize margin of decision boundary

12 Adaptive decision-making (or Random)

:D



How to “Guess” the Gradient

Observation I: If $\mathbb{E}[uu^\top] = \mathbf{I}_d$, then $\mathbb{E}[uu^\top \nabla f(x)] = \nabla f(x)$

Observation II: $\frac{f(x+\lambda u) - f(x)}{\lambda} \rightarrow u^\top \nabla f(x)$

Zeroth-order gradient estimator: sample $u \sim \mathcal{N}(0, \mathbf{I}_d)$

$$g_1(x) = \frac{f(x + \lambda u) - f(x)}{\lambda} u,$$
$$g_2(x) = \frac{f(x + \lambda u) - f(x - \lambda u)}{2\lambda} u$$

Zeroth-order optimization: at each step, $u_t \sim \mathcal{N}(0, \mathbf{I}_d)$

$$x_{t+1} \leftarrow x_t - \eta \frac{f(x_t + \lambda u_t) - f(x_t - \lambda u_t)}{2\lambda} u_t$$